Reflective Essay

Accompanying: *Prompt Cost Optimisation for Speculative Parallelism in Text-to-SQL: Schema Pruning, a Semantic Fact Store, and Cache-Stable Structure*

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1. Purpose of this essay

The research paper reports what I built and what it measured. This essay is about how that result set came to exist, which is a less orderly story than the paper implies. It covers the places where my working assumptions broke on contact with implementation, and in one case on contact with a change I did not make and could not see. It also covers what the finished work does well and badly, what a longer project would have done differently, and where the legal, social, ethical and sustainability questions sit. It is written in the first person and contains judgements about my own decisions that would be out of place in a research paper. It also records two episodes the paper mentions only as neutral methods notes. The first is a silent provider-side model replacement that invalidated a month of results. The second is a deliberate decision to narrow the project from five coordination policies to three, which cost me most of a chapter's worth of finished work and produced a better paper.

2. Theory versus practice: where the assumptions broke

The waste I set out to remove was not the waste that mattered. The project began from a visible and intuitive inefficiency. Replicas launched on the same question issue largely the same exploratory SQL, and I measured duplication rising from 39–49% at three replicas to 76–89% at twenty-five, so the obvious move was a shared cache that returns a peer's result instead of hitting the database again. It works exactly as designed and eliminates most of that traffic. It also does very little to the bill. Chat APIs are stateless, so the dominant cost is not the queries a replica issues but the prompt it re-sends on every single turn, and an instant cache hit simply frees turn budget that the replica spends exploring further, re-billing the whole context each time. Confronting that forced the central reframing of the project, which is that the object worth attacking is the prompt rather than the query stream, and that raw consumption and billed cost are two ledgers a policy can win and lose at the same time. My original hypothesis was not so much wrong as aimed at the wrong ledger, and the paper's whole spine is the correction.

Stacked measurement cannot attribute credit, and finding that out cost me two policies. For most of the project I measured layers on top of each other, adding a policy to a running stack and recording the delta, which felt efficient and was how the earlier drafts were written. It is not sound. A delta measured against a stack tells you what the layer adds to that stack and nothing about the layer, and when I finally re-ran the semantic fact store in genuine isolation, its previously reported gain disappeared, because it had been measured against a configuration that already contained the shared cache, early stopping and pruning. Repairing this properly meant re-running every arm with nothing else enabled, which was affordable only for a subset of the policies. The choice I made was to keep the three that act on the prompt and cut the two that act on the execution layer. What turned an unwelcome budget cut into the argument of the paper was noticing that the cost identity has exactly three factors, replicas times turns times the price of a prefix plus a history, and that the three surviving methods attack one factor each. The taxonomy is therefore complete over the identity rather than being a selection from a longer list, which is a stronger claim than the five-policy version could have made. I would rather have arrived at that by design than by attrition.

Prompt caching turned out to be an architectural property rather than a configuration flag. I had assumed enabling provider caching was a matter of setting an option. In fact the agent loop violated both invariants the providers require, since the fact-store injector rewrote the interior of the message list and the pruning fallback rewrote the system message in place, so no prefix was ever byte-identical to the previous turn and nothing was ever cached. The loop had to be restructured into a frozen prefix, an append-only history and an optional volatile trailer before any saving existed at all, and the byte-stability guarantee is now held in place by a unit test rather than by care. This is the method that ended up mattering most in the paper, removing 18.8–86.0% of billed input across all fifteen configurations while leaving raw consumption statistically unmoved in fourteen of them, and it is worth being honest that its size is partly a fact about price schedules rather than about my engineering. Cached input costs 50% of standard on GPT-4o mini, 10% on Gemini 2.5 Flash and 2% on DeepSeek, so the same unchanged mechanism is worth far more on one provider than another.

Pruning inverted its own goal until recall was made a hard precondition. My first schema pruner was a keyword heuristic, and it made things worse, because a task whose gold table has been dropped sends the agent exploring blind and costs more than it saved. The finished method only became reliable once hybrid scoring, foreign-key expansion and an execution-time fallback were in place, and the honest framing of the result is conditional rather than headline. Where every gold table survives, pruning removes 10–20% of raw tokens on every model at both replica counts, with all six intervals excluding zero. Where one is missing, consumption rises by 32–156%. At the 89.6% recall my pruner achieves across the full split those two regimes very nearly cancel, which makes the aggregate the least stable quantity in the whole experiment, indistinguishable from zero at three replicas and 8–11% at ten.

A provider replaced one of my models mid-experiment and my harness could not see it. On 24 July DeepSeek folded the endpoint I was using into its successor, keeping the same request format and the same registry key. My harness logged the model alias I requested rather than the identity the provider served, so nothing on disk separated results taken before the change from results taken after. It surfaced two weeks later as an anomaly I very nearly explained away, since every August run was beating every July control by six to seven points with a suspiciously sharp date boundary. Re-running the controls on the current model settled it. Seven comparisons that had shown gains between 4.8 and 7.4 points with intervals excluding zero all collapsed to null once treatment and control sat in the same model era. Those phantom effects had already passed paired bootstrap confidence intervals and independent seed replication, and the reason is that both of those check against sampling noise and neither checks against a changed world. Statistical rigour and experimental validity are separate things, which I now believe is the most transferable lesson the project taught me. On live commercial endpoints the model is an uncontrolled variable unless it is pinned, the served identity is logged per call, and any effect whose treatment and control straddle a provider change is treated as unestablished.

The subset lied to me about two of the three methods. I did most of the development on a 50-task subset for cost reasons and treated it as a small version of the truth. It is not. Two results reverse between the subset and the full 500-task split, and in both cases the subset is the one that fails. Pruning's aggregate saving of 15–29% on the subset is null at full scale, and the fact store's sign on DeepSeek flips from a 16–21% penalty to a 7% saving. What makes the second case genuinely interesting rather than merely embarrassing is that the injection load is nearly identical at both scales, 32.4 facts per task against 28.3, so the mechanism I had used to explain the penalty cannot explain its disappearance. The tax bounds the achievable saving without predicting it. This is now the reason the paper labels every 50-task magnitude as indicative only.

Refusing my own data was harder to do than to describe. After the model swap I put a rule into the analysis rather than into my judgement, so that any batch not carrying an identifier from the current experiment generation is refused rather than reused. The rule is trivial to write and genuinely unpleasant to run, because it declined a large amount of work that was very probably fine, and it left visible holes in the matrix that then had to be filled by re-running twenty-one cells I had already paid for once. The temptation to add an exception for the batches I was confident about was real, and the reason I did not is that confidence about a batch is exactly what the previous month had shown to be unreliable. A rule you can override on a case-by-case basis is not a rule, it is a preference, and preferences are what produced the phantom results in the first place.

Isolation would have told me to delete the component that pays most. Measured alone, the fact store has no reliable sign on any model, and at full scale it costs DeepSeek 3.5 points of accuracy, which is the only accuracy price anywhere in the study. An ablation stopping there deletes it. Yet the composed stack that contains it beats the multiplicative prediction from its own parts on eleven of fifteen configurations, and on DeepSeek at three replicas the parts predict a net token increase where the stack in fact saves 24%. Attribution and deployment are different questions, and I had been treating them as one.

3. Strengths of the work

The design is complete rather than opportunistic. All four arms were run across three models at three replica counts on the subset and two at full scale, giving sixty cells with no gaps to explain away, and that breadth is what caught the subset artefacts described above. Every headline comparison carries a paired bootstrap confidence interval over matched questions, and the rule that nothing crossing zero is claimed is applied uniformly, including where it costs me a result I wanted. The analysis runs in strict mode, meaning it refuses any batch not carrying a clean identifier from the current experiment generation, which deliberately shrank my own evidence base by declining legacy runs that would have been convenient to reuse. Accuracy is treated as a constraint rather than a headline, and the report of it is precise rather than flattering: fifty-eight of sixty intervals contain zero, which is fewer exceptions than sixty intervals at 95% would throw by chance, and the paper still declines to claim no effect anywhere because the two exceptions share a model, a cell and a mechanism instead of being scattered. The harness is trace-driven throughout, so every table, figure and interval regenerates from structured logs by script, and that property is the only reason a month of invalidated results could be rebuilt inside a deadline. The methods are also safe by construction rather than by tuning, since pruning always carries an execution-time fallback, the fact store is advisory and cannot alter a submitted query, and cache-stable structure changes no content byte at all.

4. Weaknesses of the work

The most expensive weakness was observability. Recording the requested alias instead of the served model identity is why a provider swap went undetected for a fortnight, and the fix is one line per API call. That was a design failure rather than bad luck. Second, and structurally more limiting, no cell in the final design is replicated. Second seeds exist only for older configurations that strict mode now refuses, so the eleven-of-fifteen composition result is a consistent direction across configurations rather than a replicated effect, and earlier work in this project saw 50-task seeds swing by up to eight accuracy points, which bounds how much weight it can carry. Third, coverage is uneven, since the twenty-five-replica column exists only on the subset and is therefore the least trustworthy part of the matrix given what the subset did to the other two methods. Fourth, the additivity check compares independent runs against a multiplicative null rather than paired ones, so no interval attaches to any gap in it and the finding is descriptive. Fifth, the 89.6% recall ceiling rests on hand-written per-database rules and foreign-key expansion gated to a single database, which is precisely the part least likely to transfer. Sixth, there is no latency evidence at all, because the batch records carry no timing field, and cost figures are list-price approximations rather than invoices. Finally, the statistical machinery arrived too late. The confidence intervals promised in the March project definition were built in August, after a draft had already gone to my supervisor carrying unqualified point estimates, several of which did not survive contact with them. Verification belonged inside the experiment loop from the first batch rather than at the end as a reporting step.

5. Work I would have done with more time

The first item is replication, because it is the single change that would most improve the paper. A second seed on every cell, or at minimum on the fifteen composed configurations, would convert the composition result from a direction into an effect. The second is a paired composed-versus-parts design run under one seed, which would let the additivity gap carry a confidence interval instead of being reported descriptively. The third is completing the twenty-five-replica column at full scale, since that column currently sits entirely on the subset that misled me twice. The fourth is a learned schema-linking stage to replace the hand-written recall rules, which is the change that would decide whether pruning generalises beyond the databases I tuned it on. The fifth is elevating the heterogeneous cross-model ensemble from a preliminary observation to a proper study. Drawing replicas from all three models reached roughly eight points over the best single model on the full split, which is the strongest hint in the entire project of where accuracy, as opposed to cost, actually lives. It is excluded from the paper because it sits on a different axis and an older configuration, and that exclusion was the right call for a cost paper, but it is the result I most regret leaving behind.

6. Further work beyond the current scope

Three directions are genuinely open. Cross-task persistence of the fact store and the cached prefix would test whether coordination value compounds across a workload rather than only within a task, and it needs an explicit re-execution gate so that a stored fact is never trusted for a new question without verification, plus a guard against benchmark leakage. Porting from SQLite to a production backend where round-trip latency and per-query billing dominate would likely invert which ledger matters, promoting the execution-layer policies this paper cut back to the headline, which would be a useful demonstration that the two-ledger framing generalises rather than that any particular ranking does. Finally, the provider-drift episode points at infrastructure work with value well beyond this project. Agent benchmark harnesses should fingerprint the served model per call and flag era boundaries automatically, so that the anomaly-hunting that caught my confound by luck becomes a property of the tooling instead of a matter of someone noticing a suspicious date.

7. Legal, social, ethical and sustainability considerations

Legal. BIRD is used under its research licence, and since no personal data or human participants were involved, ethics review was not required. All experiments ran under provider terms of service with credentials kept out of the repository. Every prompt sends schema text and evidence to an external provider, which is inconsequential for a public benchmark but would need a governance review over proprietary data. The fact store adds a second issue that is easy to miss, because it moves query-derived information between agent contexts, and in a multi-user deployment that is a data flow which must be authorised rather than assumed.

Social. The cost structure of speculative parallelism, where twenty-five replicas cost roughly thirty times a single trajectory and buy no accuracy that can be distinguished from noise, currently reserves reliable multi-replica agents for well-resourced operators. Removing most of the billed cost at unchanged accuracy lowers that barrier. The counterweight is that cheaper natural-language database access broadens who queries data directly, including people poorly placed to notice a fluent but wrong answer, so the correctness-first constraints in the design are a social safeguard as much as a technical one. Final answers always execute live and are never served from a cache or replaced by a distilled fact.

Ethical. Generative AI use is disclosed in the paper's appendix, covering implementation assistance and editorial drafting, with all quantitative content computed by deterministic scripts from execution traces, no model-generated figure or table values, and every design decision and final text reviewed by me. AI-suggested references were checked against source records before inclusion. The sharpest ethical test the project posed was self-made. When the model swap surfaced, the cross-era numbers were already written into a draft as the headline result, and they were better than the truth. Choosing to re-run and publish the smaller numbers consumed a fortnight and produced a less spectacular paper. I have come to think that building infrastructure which makes that choice cheap is itself an ethical practice, because honesty scales with how easy your tooling makes it to disbelieve yourself.

Sustainability. The waste measurements in this paper are measurements of squandered inference energy, so every accuracy-neutral saving is a Green AI result, and the largest of them removes most of the billed input across all fifteen configurations without changing a content byte. The negative findings matter just as much here. The shared cache cuts database load while raising raw token volume, which means enabling everything is not automatically greener, and the two-ledger analysis is an energy-accounting framework as much as a billing one. I should also record the cost side honestly, since verification and repair roughly doubled the project's total compute footprint. I judge that spend justified, because unverified claims propagate into other people's deployment decisions, but a better-instrumented harness that caught the confound in hours rather than weeks would have made most of it unnecessary.

8. Concluding reflection

This project changed shape three times, and each change was forced rather than chosen. The first was intellectual, moving from removing duplicated database work to identifying which cost ledger each layer actually touches, forced by a cache that eliminated most database traffic while the bill barely moved. The second was methodological, giving up stacked measurement for isolation and losing two policies in the process, which cost me finished work and produced the cleanest argument in the paper, since the three surviving methods turn out to exhaust the cost identity rather than sample it. The third came from outside, when a provider replaced one of my three models mid-experiment and the phantom gains it manufactured passed straight through the statistical machinery I had just built. Detecting that, accepting it, and re-basing every affected result in the closing weeks was the hardest thing that happened here and the most valuable. The paper that survived is less exciting than the drafts before it. Its accuracy story is a disciplined null with two declared exceptions, its savings are real but specific to the ledger that binds, and its largest single number belongs to a provider's price schedule rather than to a clever algorithm. I am more confident defending it than any earlier version, and the reason generalises past this dissertation. The results I can trust are exactly the ones my infrastructure made cheap to re-derive, re-run and doubt.